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PERSONALIZED FLUID AND FUEL INTAKE FOR PERFORMANCE OPTIMIZATION IN THE HEAT

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ABSTRACT

It is well appreciated that a loss of body water (dehydration) can impair endurance performance and that the effect is magnified by environmental heat stress. A majority of professional sports medicine and nutrition organizations recommend drinking during exercise to replace sweat losses and prevent dehydration, while also avoiding frank over-hydration. Knowledge of sweating rate, which is highest in the heat for any given metabolic rate, is therefore considered key to developing a sound drinking strategy. Exercise duration and the provision of liquid fuel interacts with required drink volumes in important ways that are infrequently discussed but are of utmost practical concern. This review details some challenges related to the optimized coupling of fluid and fuel needs during prolonged exercise in the heat and the need for personalization.

Key words: dehydration, hyponatremia, exercise fluid replacement, personalized nutrition

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PERSONALIZED FLUID AND FUEL NEEDS FOR EXERCISE IN THE HEAT

When sweat losses exceed fluid intakes during exercise, body water deficits accumulate (dehydration), cardiovascular strain is measurably increased, and endurance exercise performance is frequently impaired, particularly when accompanied by environmental heat stress[1-6]. The physiological mechanisms to explain endurance performance impairments have been well studied but are not yet precisely resolved[7-9]. Knowledge of sweating rate, which is highest in the heat for any given metabolic rate[10], is key to developing a sound drinking strategy that minimizes dehydration while also guarding against overdrinking. The 2007 position stand of the American College of Sports Medicine[11] was the first to advocate for drinking rates precisely high enough to prevent meaningful dehydration (> 2% body mass), but low enough to prevent over-hydration. In other words, the 2007 ACSM position stand recommends a personalized exercise fluid replacement strategy based on individual sweating rates or sweat losses. Since its publication, many professional sports medicine organizations have recommended the same, including the Academy of Nutrition and Dietetics, the Canadian Dietetics Association, the National Athletic Trainer's Association, the International Amateur Athletic Federation, and the International Olympic Committee, among others. However, personalized drinking strategies depend importantly on an athlete's personal goals, as well as the rules and conventions of a particular sport which dictate the availability of fluid and fuel in competition[12-17]. The provision of fuel[18] interacts with required drink volumes in important ways that are infrequently discussed but are of utmost practical concern. This review details some challenges related to the optimized coupling of fluid and fuel needs during prolonged exercise in the heat and the need for personalization.

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PERSONALIZED FLUID INTAKE USING SWEAT RATE

The analyses presented in this review are predicated upon adopting a meaningful dehydration avoidance threshold of > 2% body mass (i.e., > 2% loss of body water expressed as a percentage of mass)[8, 11], but we acknowledge the potential for variation around this threshold[9]. The recommendation to prevent dehydration during exercise[11] requires precise knowledge of sweating rate. In this paper, we examine the impact of anticipated, sustainable sweating rates of 0.5, 1.0, and 1.5 L/h on dehydration and required drinking rates. This range covers a realistic majority of athletes [19, 20] and is almost exactly the range reported by Byrne et al. [21] for runners during a warm and humid half marathon. We use examples of both smaller (50 kg) and larger (90 kg) runners to illustrate the impact of body size, since dehydration is expressed as a percentage of body mass.

In hot environments, dry heat loss may become limited and there can even be dry heat gain. In these circumstances, body cooling becomes increasingly dependent on evaporative sweat loss. The complete evaporation of 0.5, 1.0, and 1.5 L/h of sweat can remove 337, 675, and 1012 W of heat energy, respectively[22]. For the same running distance and running duration, a larger person will require a larger sweating rate (all other factors equal). For example, we can calculate using 1 kcal/kg/km[23] that running 21 km in 2 hours will produce 611 W of heat for a 50 kg runner, which would require the evaporation of ~ 0.9 L/h of sweat to balance. The same effort for a 90 kg runner would produce 1100 W of heat and require the evaporation of ~1.6 L/h of sweat to balance. However, if the larger runner takes 2.5 hours, the required evaporative sweat rate reduces to ~0.95 L/h. Similarly, if the 50 kg runner was an elite runner who completes the distance in 1 hour, the requirement for evaporative cooling would increase to a sweating rate of

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~1.8 L/h. Therefore, the range of sweating rates evaluated is plausible for the range of body masses evaluated, depending on exercise intensity, environment, and evaporative efficiency[10]. Although dehydration takes time to develop for all, it will occur faster in smaller runners than in larger runners when we assume sweating rates are identical for both. Therefore, exercise duration is also critical.

The sweating rates of 0.5, 1.0, and 1.5 L/h were accumulated for 1 h, 2 h, and 4 h durations of exercise to coincide with distinct recommendations for fuel rate consumption (discussed below)[18]. The sum of accumulated sweat losses ranged from 0.5 to 1.5 L for 1 hour of exercise, 1.0 to 3.0 L for 2 hours of exercise, and 3.0 to 6.0 L for 4 hours of exercise, respectively. Figure 1 illustrates how much dehydration would accumulate by the end of exercise if choosing to drink nothing. The top of each bar is the dehydration level for a 90 kg runner and the bottom a 50 kg runner. Runners measuring between 50 and 90 kg fall between the top and bottom of each floating bar. In 11 out of 18 scenarios, dehydration would reach > 2% of body mass, but in 7 out of 18 it would not. If meaningful dehydration avoidance is the only concern, drinking would not be necessary for 1 hour of exercise unless a runner was ≤ 70 kg and had a sweating rate of ~1.5 L/h. For 2 h of exercise, sweating rates ≤ 0.5 L/h for any size runner require no drinking. In all other scenarios depicted, some level of drinking would be necessary to prevent meaningful dehydration. Importantly, even in the above scenarios where no drinking is required 'during exercise,' knowledge of individual sweating rate remains important for gauging replacement needs in the post-exercise period[9].

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Figure 2 A and B depict the drinking rates required to prevent > 2% dehydration by the end of exercise for 50 kg (A) and 90 kg (B) runners. For smaller runners, high sweating rates (1.5 L/h) require drinking rates of 0.5 to 1.25 L/h (1 to 4 h exercise), moderate sweating rates (1.0 L/h) require drinking rates of 0.5 to 0.75 L/h (2 to 4 h exercise), and low sweating rates (0.5 L/h) require drinking rates of just 0.25 L/h, but only when exercise will last 4 h. Larger runners (90 kg) require minimal (< 0.25 L/h) drinking when sweating rate is 0.5 L/h, independent of exercise duration (up to 4 hours). When sweating rate is 1.0 L/h, some drinking (≤ 0.5 L/h) is required when exercise duration is 2 to 4 h, but more substantial drinking volumes are required for the same durations when sweating rate is 1.5 L/h (0.5 to 1.0 L/h).

TRADEOFFS WITH PERSONALIZED FUEL INTAKE

The tradeoff between fluid and fuel intake during endurance exercise has been intermittently discussed for 30 years[2, 24-27]. Over the past 5 years, fuel intake recommendations for competitive endurance athletes have been updated to include the importance of exercise duration. Recommendations by Jeukendrup (2014)[18] have been adopted by sports medicine and sports dietetics organizations (e.g., Thomas et al., 2016)[28] worldwide. It is recommended that up to 30 g, 60 g, and 90 g of carbohydrates (CHO) be ingested when exercise is 1 to 2 h, 2 to 3 h, and > 2.5 h duration. We selected 1 h, 2 h, and 4 h exercise durations to represent running distances of 10 km, 21 km, and 42 km based on median finishing times for large road races represented by these distances. The consumption of a typical sports drink will deliver ~60 g/h CHO in a 1 L volume of fluid. Figure 3 A and B depict how much fuel (g/h CHO) is obtained when consuming a sports drink at rates designed to prevent meaningful dehydration (Figure 2 A/B). It

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is clear from Figures 3 A and B that fuel intake goals will rarely be met when drinking in accordance with fluid needs.

What would happen if a choice was made to use sports drink to meet fuel needs, independent of fluid needs? Figure 4 shows the expected changes in hydration status (±% body mass) when drinking to meet fuel needs using a conventional sports drink. For example, when matching sweating and drinking rates of 0.5 L/h, 1.0 L/h, and 1.5 L/h, 30, 60, and 90 g/h CHO are obtained when consuming a typical sports drink and there is no change in body mass. Since 0.5 L/h of sports drink provides 30 g/h of CHO fuel, sweating rates of 1.0 to 1.5 L/h result in mild dehydration when exercise is 1 h duration. When exercise is 2 h duration, a gain or loss of 2% body mass results for 0.5 and 1.5 L/h sweating rates, respectively. When exercise is 4 h duration, the desire to meet 90 g/h fuel needs requires substantial over-drinking and over-hydration if CHO is obtained using sports drink only. A single sports drink formulation delivering ~60 g/L CHO, though perhaps optimal in many respects, cannot meet the combined heterogeneous fluid and energy needs of all individuals.

PERSONALIZED FLUID INTAKE USING THIRST

An alternative to personalizing exercise fluid replacement based on knowledge of individual sweating is to use a different form of personalization based on individual thirst. Drinking to thirst has been recommended specifically as a means of reducing the risk of hyponatremia during prolonged exercise[29] largely on the basis that during heavy exertion, thirst lags behind sweat losses, making overdrinking less likely. However, it should be noted that there are reported incidents of individuals becoming hyponatremic when consuming fluids according to thirst[29]. As

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a strategy for satisfying fluid needs for exercise of any duration this approach is highly contingent[27] and to our knowledge has only twice[27, 30] been considered specifically in light of the need for satisfying both fluid and fuel needs.

Any investigation of the hypothetical success of a drink to thirst strategy in this scenario of a fluid and fuel tradeoff requires an estimate of thirst-driven drink volumes, which we have set to 50% of sweating rate for this exercise[31, 32]. It should be noted that 50% replacement of fluid losses is at the higher end of thirst-driven fluid intake (*ad libitum*) and others have reported lower intakes (~20%)[33, 34]. This makes comparison to Figures 2 and 3 straightforward as drink volumes are made to be precisely 0.25, 0.5, and 0.75 L/h and CHO intakes 15, 30, and 45 g/h when consuming a conventional sports drink, respectively. Of the 9 possible scenarios each for 50 kg (Figure 2A) and 90 kg (Figure 2B) runners, too much or too little drinking would occur, relative to needs, in 14/18 cases. Importantly, overdrinking relative to sweating rate would never occur (using 50% replacement via thirst). For fuel (Figures 3A and B), it is the same, thus in 28/36 scenarios some combination of fluid or fuel is not met (too much or too little). For example, a 50 kg runner with a 1.5 L/h sweating rate exercising for 2 hours would require 1 L/h of fluid and 60 g/h CHO (Figure 3 and 4A), but would receive only 0.75 L/h and 45 g/h CHO using thirst. If the runner was 90 kg with a 1.5 L/h sweating rate exercising for 1 hour, no fluid is required but 30 g/h CHO is still recommended. In this example, drinking to thirst would provide more fluid (0.75 L/h) and fuel (45 g/h) than needed.

CONCLUSIONS

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Exercise fluid and fuel needs depend heavily on exercise intensity, environment, and exercise duration, as outlined above. Our analysis indicates that knowledge of individual sweating rates can be used to closely gauge drinking needs for optimizing fluid replacement[11]. However, the trade-off between fluid and fuel will often require use of food products other than sports drinks to satisfy energy needs[18] and avoid over-drinking when exercise is ≥ 2 hours (Figure 4). More concentrated blends of cleverly formulated drinks, gels[35], or simple use of easily digestible CHO food sources may be required and should be tested for individual preference and tolerance. In our analysis above, a drink to thirst strategy sometimes resulted in consumption of more, sometimes less, fluid and energy than needed for the circumstances. Better or worse outcomes are plausible but depend heavily on the precise proportion of sweat losses replaced using individual thirst sensation. Thoughtful, personalized fluid intake strategies are needed for optimizing exercise performance in the heat.

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Figure 1. Impact of accumulating sweat losses on the development of dehydration in the absence of drinking. Floating bars represent minimum (top of bar, 90 kg runners) to maximum (bottom of bar, 50 kg runners). The dashed lines represent the threshold for meaningful dehydration (> 2% body mass). Colored bars denote respective sweating rates.

Figure 1

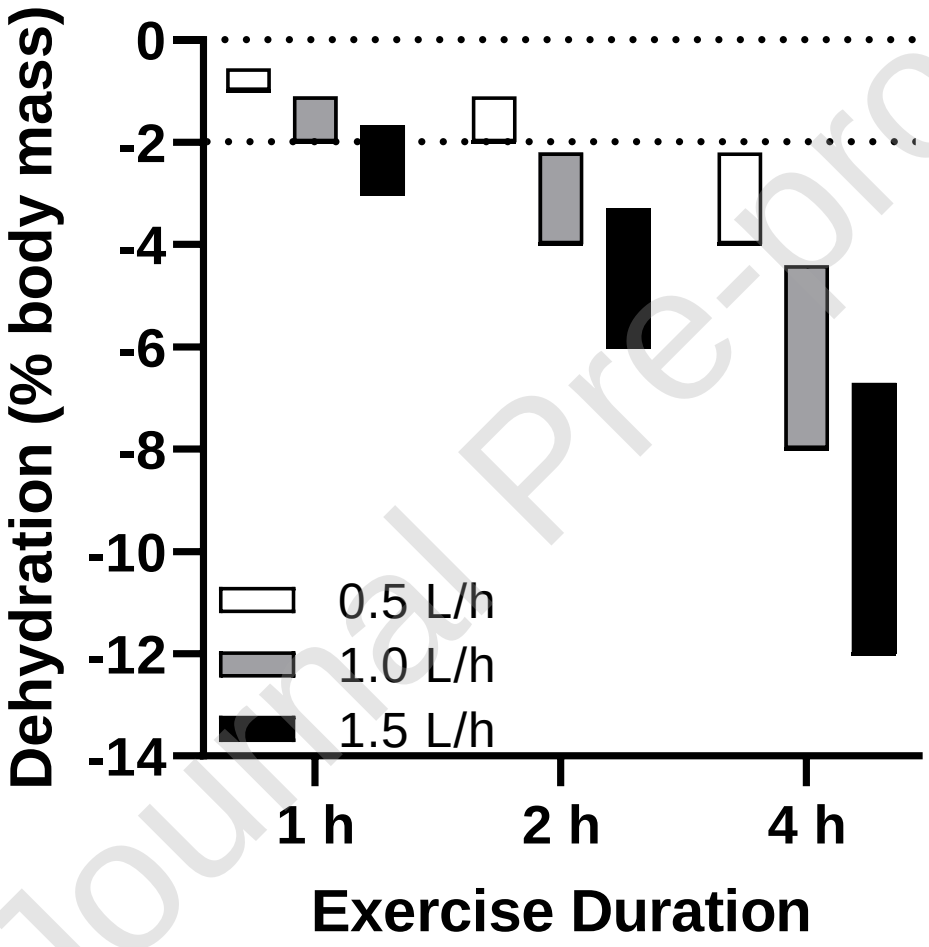

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Figure 2. Required hourly drinking rates to prevent meaningful dehydration (> 2% body mass) for 50 kg (A) and 90 kg (B) runners. The absence of columns is synonymous with no requirement for drinking. Colored bars denote respective sweating rates.

Figure 2

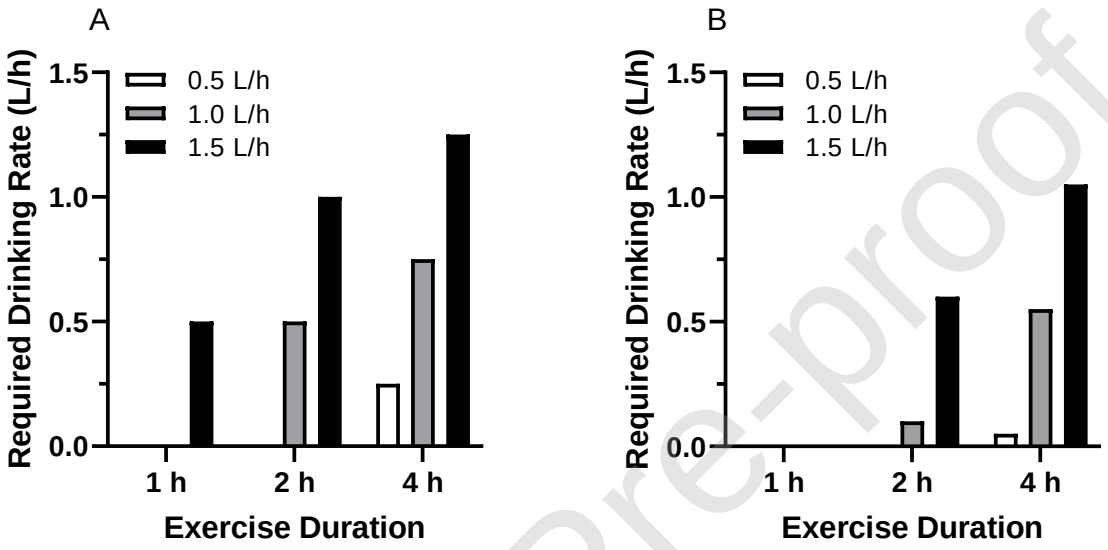

Figure 3. Hourly fuel consumption achieved when consuming a typical sports drink at rates to prevent meaningful dehydration (> 2% body mass) for 50 kg (A) and 90 kg (B) runners. The absence of columns is synonymous with no fuel due to absent drinking. Colored bars denote respective sweating rates. Dashed lines at 30, 60, and 90 g represent fuel intake recommendations for 1h, 2h, and 4h of exercise, respectively.

Figure 3

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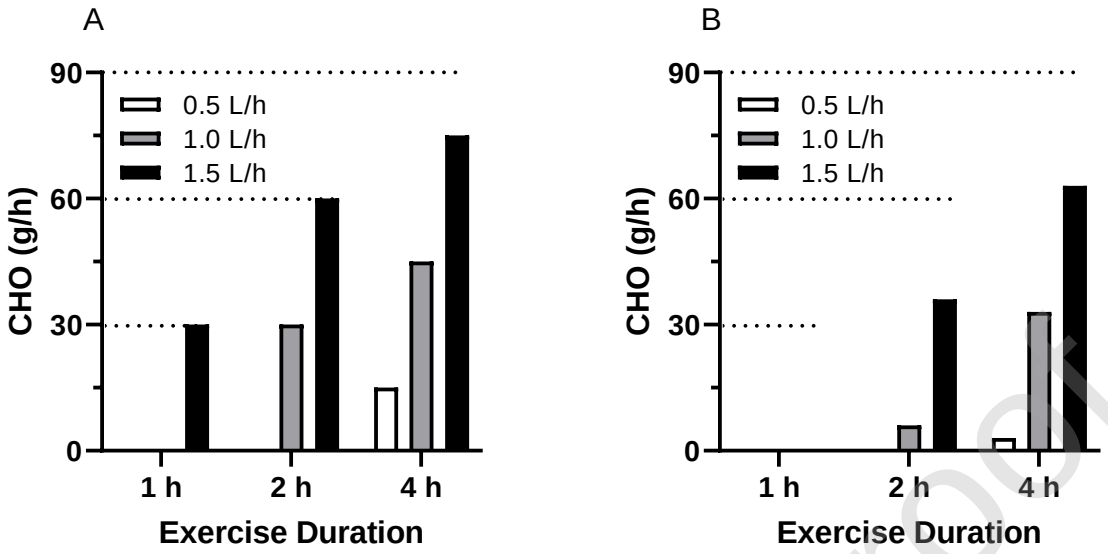

Figure 4. Impact on hydration status when using a typical sports drink to achieve fuel intake goals. Floating bars represent minimum to maximum. Values below zero represent dehydration whereby the top of the bar is for 90 kg runners and the bottom is for 50 kg runners; values above zero represent over-hydration whereby the top of the bar is for 50 kg runners and the bottom is for 90 kg runners. The dashed lines represent the threshold for meaningful dehydration (> 2% body mass). Colored bars denote respective sweating rates.

Figure 4

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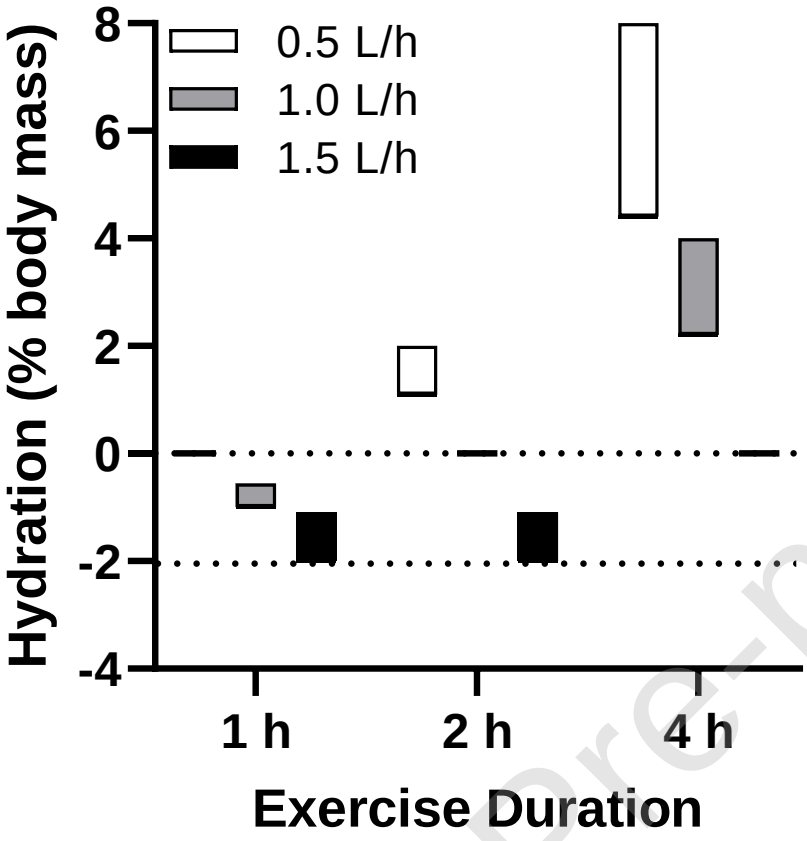

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